Planning and Reinforcement Learning in AI Systems

(PARLAI)

Lesson 9b - Model-based RL (part 2)

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Recap

In the previous module, we introduced **model-based RL**:

experience $\longrightarrow$ learn model $\longrightarrow$ plan $\longrightarrow$ act

The learned model predicts consequences of actions:

$$(s_t, a_t) \longrightarrow (s_{t+1}, r_{t+1}).$$

Current focus: How to use an imperfect model safely and efficiently?

Model Error and Compounding Errors

Short-Horizon Planning and Replanning

Dyna and Simulated Experience

Estimating Model Uncertainty

Conclusion

How can an agent make good decisions using a model it does not fully trust?

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Advanced model-based RL combines:

- model learning
- uncertainty estimation
- planning
- exploration

Model Error and Compounding Errors

One-Step Model Error

Suppose the true dynamics are

$$s_{t+1} \sim p(s_{t+1} \mid s_t, a_t),$$

and the learned model is

$$\hat{s}_{t+1} = f_\eta(s_t, a_t).$$

A natural one-step error measure is:

$$\epsilon_t = \|s_{t+1} - f_\eta(s_t, a_t)\|$$

When is this measure relevant?

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One-Step Model Error

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When is this measure relevant?

The assumption is that the state is represented as a vector in a metric space, so that subtraction and a norm are meaningful.

Compounding Model Error

A small one-step error is useful, but planning usually depends on many predicted steps.

A model rollout recursively feeds predictions back into the model:

$$\hat{s}_{t+1} = f_\eta(s_t, a_t),$$

$$\hat{s}_{t+2} = f_\eta(\hat{s}_{t+1}, a_{t+1}),$$

$$\hat{s}_{t+3} = f_\eta(\hat{s}_{t+2}, a_{t+2}).$$

Even if each one-step error is small, the rollout can drift away from the true trajectory.

Small local errors can lead to large long-horizon planning errors.

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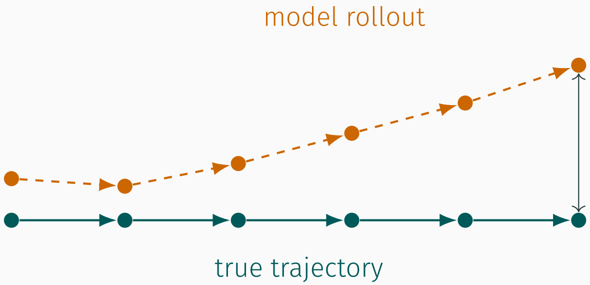
Compounding Model Error



The farther we roll out the model,
the more cautious we should be.

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the more cautious we should be.

Distribution Shift

The model is trained on data collected from previous behavior:

$$(s, a) \sim d_{\text{data}}(s, a).$$

where d_{data} is the state-action visitation distribution induced by the behavior policy that generated the data.

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Distribution Shift

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The model is trained on data collected from previous behavior:

$$(s, a) \sim d_{\text{data}}(s, a).$$

where d_{data} is the state-action visitation distribution induced by the behavior policy that generated the data.

But planning may ask the model about new regions:

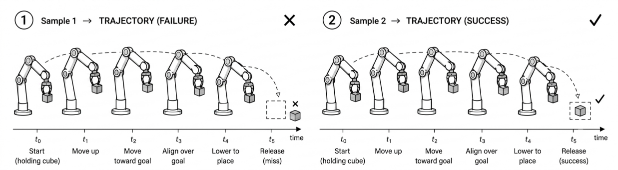
$$(s, a) \sim d_{\text{plan}}(s, a).$$

If $d_{\text{plan}}(s, a) \neq d_{\text{data}}(s, a)$, then the model may be used in parts of the state-action space where it has little evidence.

A model is most reliable near the data that trained it.

Where have we already encountered this issue?

Example: Learned Manipulation Dynamics



Suppose the robot learned a model for moving a cube:

$(q_t, a_t) \longrightarrow q_{t+1}$, where q_t is the robot configuration.

The one-step prediction may be accurate $\hat{q}_{t+1} \approx q_{t+1}$, but after many imagined actions:

- the gripper may drift away from the cube,
- contact may be predicted incorrectly,
- a collision may be missed,
- the final predicted grasp may be infeasible.

In robotics, long open-loop model rollouts can be especially fragile.

Addressing Model Error

Ideas?

Model Error and Compounding Errors

Short-Horizon Planning and Replanning

Dyna and Simulated Experience

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Ideas?

- **Short-horizon planning and replanning** - Use the model only for a few steps, execute one action, then replan.
- **Using the acquired model to learn faster** - Generate simulated experience to update the value function or policy.
- **Estimate model uncertainty** - Detect where the learned model is reliable and where it is uncertain.
  - **Explore to improve the model** - Choose actions that reduce uncertainty or reveal useful new information.
  - **Plan conservatively when needed** - Penalize risky or uncertain trajectories in safety-critical domains.

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Short-Horizon Planning and Replanning

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If long model rollouts are unreliable, one solution is to use the model only for a short horizon.

Instead of planning an entire episode, solve:

$$a_t, \ldots, a_{t+H} = \arg\max_{a_t, \ldots, a_{t+H}} \mathbb{E}_{\hat{M}} \left[\sum_{k=0}^{H} \gamma^k r_{t+k+1} \right].$$

Then execute only the first action a_t, observe the real next state, and replan.

Short-horizon planning limits how far we trust the model.

This is often called Model Predictive Control (MPC), which repeatedly plans over a finite horizon.

Short-Horizon Planning and Replanning

MPC with a Learned Model

1. Given current state s_t, sample or optimize candidate action sequences:
$$a^{(1)}_{t:t+H}, \ldots, a^{(N)}_{t:t+H}$$
2. Roll each sequence forward using the learned model $\hat{M}$
3. Estimate the return of each candidate sequence
4. Select the first action of the best sequence: $a_t = a_t^{(i^*)}$.
5. Execute a_t in the real environment

Replanning lets the real environment correct the model's predictions after every step.

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A simple model-based planner is random shooting.

Sample candidate action sequences:

$$a_{t:t+H}^{(i)} \sim \text{proposal distribution}, \quad i = 1, \ldots, N.$$

Evaluate each sequence using the learned model:

$$\hat{G}^{(i)} = \sum_{k=0}^{H} \gamma^k \hat{r}_{t+k+1}^{(i)}.$$

Choose the best first action:

$$a_t = a_t^{(i^*)}, \qquad i^* = \arg\max_i \hat{G}^{(i)}.$$

How to choose H?

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Dyna and Simulated Experience

Use the Model to Expedite Learning

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Another way to use a learned model is for generating additional training data.

Real experience is expensive:

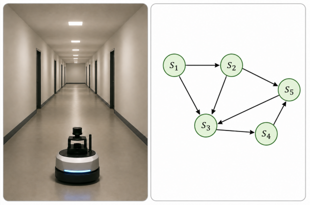
$$(s, a, r, s') \sim \text{environment.}$$

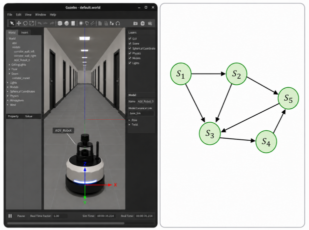
Simulated experience (i.e., data generated from the learned model) is cheaper:

$$(\tilde{s}, \tilde{a}, \tilde{r}, \tilde{s}') \sim \hat{M}.$$

Use the Model to Expedite Learning

Real experience vs. dreamed experiences





Note that 'real experience' can be extracted from a simulator

Dyna-Style Learning

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Dyna uses both real and 'imagined' (model-generated) experience

$$\hat{s}' \sim \mathcal{P}_\eta(\cdot \mid s, a), \qquad \hat{r} \sim \mathcal{R}_\eta(\cdot \mid s, a).$$

Imagined transitions of the learned model serve as a source of additional training experience.

$$s \sim \mathcal{P}(\cdot \mid s, a), \qquad r \sim \mathcal{R}(\cdot \mid s, a).$$

Real transitions are used for:

1. updating the value function or policy
2. updating the learned model which is used to generate additional imagined transitions:

Key Insight: The same learning rule can be applied to real and model-generated transitions.

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Key Insight: The same learning rule can be applied to real and model-generated transitions.

Suppose the agent maintains a Q-function. For a real transition, it performs the usual Q-learning update:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right]$$

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Then it updates the model:

$$\hat{M}(s, a) \leftarrow (r, s').$$

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Dyna-Q

Key Insight: The same learning rule can be applied to real and model-generated transitions.

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$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right]$$

Then it updates the model:

$$\hat{M}(s, a) \leftarrow (r, s').$$

Then it samples imagined transitions from the model and applies the same Q-learning update again.

$$\hat{s}' \sim \mathcal{P}_\eta(\cdot \mid s, a), \qquad \hat{r} \sim \mathcal{R}_\eta(\cdot \mid s, a).$$

Each real interaction can produce many additional learning updates.

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Dyna-Q Algorithm

Dyna-Q

1. Initialize $Q(s, a)$ and model $\hat{M}$.
2. Repeat:
   1. choose action a using an exploration policy,
   2. execute a, observe r, s',
   3. update $Q(s, a)$ from the **real transition** a, r, s',
   4. update the model $\hat{M}(s, a)$,
   5. for k steps:
      1. sample a previously observed pair $(\tilde{s}, \tilde{a})$,
      2. **simulate** $(\tilde{r}, \tilde{s}') \sim \hat{M}(\tilde{s}, \tilde{a})$,
      3. update $Q(\tilde{s}, \tilde{a})$.

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Why Is This Useful?

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Model-free RL learns only from real interactions.

$$\text{real experience} \quad \longrightarrow \quad \text{value/policy update}$$

Dyna-style RL learns from both real and imagined (model-based) experience.

$$\text{real experience} \quad \longrightarrow \quad \text{learn model} \quad \longrightarrow \quad \text{many imagined updates}$$

The model increases the amount of learning that can be extracted from each real interaction.

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Dyna-style RL learns from both real and imagined (model-based) experience.

$$\text{real experience} \quad \longrightarrow \quad \text{learn model} \quad \longrightarrow \quad \text{many imagined updates}$$

The model increases the amount of learning that can be extracted from each real interaction.

This can improve sample efficiency, especially when real-world interaction is expensive.

Caveats?

How to Choose Useful Simulated Updates

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Dyna raises a new question:

Which simulated experiences should we generate?

How to Choose Useful Simulated Updates

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Dyna raises a new question:

Which simulated experiences should we generate?

Options:

- sample uniformly from previously visited state-action pairs,
- focus on states where values recently changed,
- focus on uncertain parts of the model,
- focus on states likely to affect the current policy.

Example Approach: Prioritized Sweeping

Prioritized sweeping focuses planning updates where they are most useful.

If the value of a state changes a lot, then predecessor states may also need updates.

$$\Delta(s) = |V_{\text{new}}(s) - V_{\text{old}}(s)|.$$

Large $\Delta(s)$ means:

- the value estimate changed significantly,
- predecessor states may now have outdated values,
- planning should update them soon.

Prioritized sweeping uses the model to propagate important value changes backward.

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Bounding Value Error

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A learned model may have small one-step errors.

$$\hat{P}(\cdot|s,a) \approx P(\cdot|s,a), \qquad \hat{R}(s,a) \approx R(s,a)$$

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A learned model may have small one-step errors.

$$\hat{P}(\cdot|s,a) \approx P(\cdot|s,a), \qquad \hat{R}(s,a) \approx R(s,a)$$

But planning uses the model repeatedly over many future steps.

The Simulation Lemma bounds the effect of one-step model errors on long-term value estimates.

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Theorem (Simulation Lemma)

Let $M = \langle \mathcal{S}, \mathcal{A}, P, R \rangle$ be the true model, and let $\hat{M} = \langle \mathcal{S}, \mathcal{A}, \hat{P}, \hat{R} \rangle$ be a learned model.
Assume that:

- *for all s, a:*

$$|R(s,a) - \hat{R}(s,a)| \leq \epsilon_R$$

  and

$$\|P(\cdot|s,a) - \hat{P}(\cdot|s,a)\|_1 \leq \epsilon_P.$$

- *rewards are bounded by $R_{\max}$*

then for any fixed policy π,

$$\|V_M^\pi - V_{\hat{M}}^\pi\|_\infty \leq \frac{\epsilon_R}{1-\gamma} + \frac{\gamma R_{\max} \epsilon_P}{(1-\gamma)^2}$$

From the Bellman equations, for a fixed policy π:

$$V_M^\pi(s) = R^\pi(s) + \gamma \sum_{s'} P^\pi(s'|s) V_M^\pi(s')$$

and

$$V_{\hat{M}}^\pi(s) = \hat{R}^\pi(s) + \gamma \sum_{s'} \hat{P}^\pi(s'|s) V_{\hat{M}}^\pi(s').$$

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From the Bellman equations, for a fixed policy π:

$$V^\pi_M(s) = R^\pi(s) + \gamma \sum_{s'} P^\pi(s'|s) V^\pi_M(s')$$

and

$$V^\pi_{\hat{M}}(s) = \hat{R}^\pi(s) + \gamma \sum_{s'} \hat{P}^\pi(s'|s) V^\pi_{\hat{M}}(s').$$

We want to bound:

$$\Delta = \|V^\pi_M - V^\pi_{\hat{M}}\|_\infty.$$

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Separate the Error Terms: For any state s,

$$|V_M^\pi(s) - V_{\hat{M}}^\pi(s)| \le |R^\pi(s) - \hat{R}^\pi(s)| \\ + \gamma \left| \sum_{s'} P^\pi(s'|s) V_M^\pi(s') - \sum_{s'} \hat{P}^\pi(s'|s) V_{\hat{M}}^\pi(s') \right|.$$

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Bound the Transition Error: after adding and subtracting $\sum_{s'} \hat{P}^\pi(s'|s) V_M^\pi(s')$, we get two terms:

$$
\begin{aligned}
|V_M^\pi(s) - V_{\hat{M}}^\pi(s)| \leq \epsilon_R \\
&+ \gamma \left| \sum_{s'} \left(P^\pi(s'|s) - \hat{P}^\pi(s'|s) \right) V_M^\pi(s') \right| \\
&+ \gamma \left| \sum_{s'} \hat{P}^\pi(s'|s) \left(V_M^\pi(s') - V_{\hat{M}}^\pi(s') \right) \right| .
\end{aligned}
$$

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Bound the Transition Error: after adding and subtracting $\sum_{s'} \hat{P}^\pi(s'|s)V_M^\pi(s')$, we get two terms:

$$\begin{aligned} |V_M^\pi(s) - V_{\hat{M}}^\pi(s)| \leq \epsilon_R \\ &+ \gamma \left| \sum_{s'} \left(P^\pi(s'|s) - \hat{P}^\pi(s'|s) \right) V_M^\pi(s') \right| \\ &+ \gamma \left| \sum_{s'} \hat{P}^\pi(s'|s) \left(V_M^\pi(s') - V_{\hat{M}}^\pi(s') \right) \right|. \end{aligned}$$

Using

$$\|V_M^\pi\|_\infty \leq \frac{R_{\max}}{1-\gamma},$$

we obtain:

$$|V_M^\pi(s) - V_{\hat{M}}^\pi(s)| \leq \epsilon_R + \gamma \epsilon_P \frac{R_{\max}}{1-\gamma} + \gamma \Delta.$$

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Solve the Inequality: Taking the maximum over states gives

$$\Delta \leq \epsilon_R + \gamma \epsilon_P \frac{R_{\max}}{1-\gamma} + \gamma \Delta.$$

Bounding Value Error

Solve the Inequality: Taking the maximum over states gives

$$\Delta \leq \epsilon_R + \gamma \epsilon_P \frac{R_{\max}}{1-\gamma} + \gamma \Delta.$$

Move $\gamma\Delta$ to the left:

$$(1-\gamma)\Delta \leq \epsilon_R + \gamma \epsilon_P \frac{R_{\max}}{1-\gamma}.$$

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Solve the Inequality: Taking the maximum over states gives

$$\Delta \leq \epsilon_R + \gamma \epsilon_P \frac{R_{\max}}{1-\gamma} + \gamma \Delta.$$

Move $\gamma\Delta$ to the left:

$$(1-\gamma)\Delta \leq \epsilon_R + \gamma \epsilon_P \frac{R_{\max}}{1-\gamma}.$$

Therefore:

$$\Delta \leq \frac{\epsilon_R}{1-\gamma} + \frac{\gamma R_{\max} \epsilon_P}{(1-\gamma)^2}.$$

Bounding Value Error

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What does this reveal?

$$\Delta \leq \frac{\epsilon_R}{1-\gamma} + \frac{\gamma R_{\max} \epsilon_P}{(1-\gamma)^2}.$$

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What does this reveal?

$$\Delta \leq \frac{\epsilon_R}{1-\gamma} + \frac{\gamma R_{\max} \epsilon_P}{(1-\gamma)^2}.$$

Small one-step model errors can become large long-horizon value errors.

The bound contains the term $\frac{1}{(1-\gamma)^2}$

When γ is close to 1, the effective planning horizon is long, and even small transition errors can strongly affect value estimates.

This motivates short-horizon planning, replanning, uncertainty-aware models, and conservative use of learned rollouts.

Estimating Model Uncertainty

Estimating Model Uncertainty

A third approach to dealing with model error is to estimate how uncertain the learned model is.

This uncertainty estimate can be used in two complementary ways:

- **Exploration:** collect more data in state-action regions where the model is uncertain, especially in areas where additional data is expected to improve performance.
- **Robustness:** choose actions that perform well even when the learned model may be inaccurate.

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Why Model Uncertainty Matters

Knowing the model's uncertainty can be as important as knowing its prediction.

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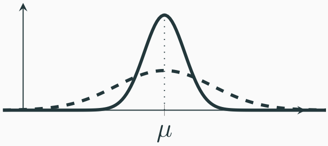
Knowing the model's uncertainty can be as important as knowing its prediction.

A learned model may make two predictions with the same mean:

$$\mathbb{E}[s_{t+1} \mid s^{(1)}, a^{(1)}] = \mathbb{E}[s_{t+1} \mid s^{(2)}, a^{(2)}].$$

But the reliability of the two predictions may be very different. For example, the model may predict either $p_\eta(s_{t+1} \mid s^{(1)}, a^{(1)}) = \mathcal{N}(\mu, \sigma_1^2)$ or $p_\eta(s_{t+1} \mid s^{(2)}, a^{(2)}) = \mathcal{N}(\mu, \sigma_2^2)$, such that $\sigma_1^2 < \sigma_2^2$.



Both predictions have the same mean, but one is more confident.

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Reminder: Aleatoric and Epistemic Uncertainty

Aleatoric uncertainty comes from randomness in the environment, i.e., $s_{t+1} \sim p(s_{t+1} \mid s_t, a_t)$.

Even with infinite data, the outcome may remain stochastic.

Epistemic uncertainty comes from lack of knowledge.

The model is uncertain because it has not seen enough data near (s, a).

Aleatoric uncertainty is part of the world.
Epistemic uncertainty can be reduced by exploration.

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Once the model produces a prediction, we still need to answer two questions.

1. **How uncertain is the model?**

$$U_\eta(s,a) \approx \text{uncertainty of } p_\eta(s_{t+1} \mid s,a).$$

2. **How should the agent account for this uncertainty?**

$$a_t = \arg\max_a \text{uncertainty-aware objective ?}.$$

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In some cases, we aim to predict a single value or state.

In others, the environment itself is uncertain. Thus, instead of predicting a single next state,

$$\hat{s}_{t+1} = f_\eta(s_t, a_t),$$

we predict a distribution:

$$s_{t+1} \sim p_\eta(s_{t+1} \mid s_t, a_t).$$

For example, a Gaussian model predicts:

$$p_\eta(s_{t+1} \mid s_t, a_t) = \mathcal{N}\left(\mu_\eta(s_t, a_t), \Sigma_\eta(s_t, a_t)\right).$$

Aleatoric or epistemic uncertainty?

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Estimating Uncertainty with Probabilistic Models

A probabilistic dynamics model predicts a distribution:

$$s_{t+1} \sim p_\eta(s_{t+1} \mid s_t, a_t).$$

For example, a Gaussian model predicts

$$p_\eta(s_{t+1} \mid s_t, a_t) = \mathcal{N}\left(\mu_\eta(s_t, a_t), \Sigma_\eta(s_t, a_t)\right).$$

Large predictive variance means the model expects many possible next states.

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Epistemic Uncertainty with Model Ensembles

A common way to estimate epistemic uncertainty is to train several models: $p_{\eta_1}, p_{\eta_2}, \ldots, p_{\eta_K}$.

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Epistemic Uncertainty with Model Ensembles

A common way to estimate epistemic uncertainty is to train several models: $p_{\eta_1}, p_{\eta_2}, \ldots, p_{\eta_K}$.

Each model is trained on the same data, but may generalize differently.

For a given state-action pair, each model predicts

$$\mu_k(s,a) = \mathbb{E}_{p_{\eta_k}}\left[s_{t+1} \mid s,a\right].$$

The ensemble mean is $\bar{\mu}(s,a) = \frac{1}{K}\sum_{k=1}^{K} \mu_k(s,a)$

Model disagreement estimates epistemic uncertainty:

$$U_{\text{epi}}(s,a) = \frac{1}{K}\sum_{k=1}^{K} \left\|\mu_k(s,a) - \bar{\mu}(s,a)\right\|^2$$

Disagreement means data does not strongly determine prediction.

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Once we have an uncertainty estimate $U_\eta(s, a)$, how can we use it?

We can use it in different ways.

Exploration: encourage actions where the model is uncertain:

$$r_{\text{explore}}(s, a) = r(s, a) + \beta U_\eta(s, a).$$

This pushes the agent toward state-action regions where more data may be useful.

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Model Uncertainty

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Exploration: encourage actions where the model is uncertain:

$$r_{\text{explore}}(s, a) = r(s, a) + \beta U_\eta(s, a).$$

This pushes the agent toward state-action regions where more data may be useful.

Robustness: penalize actions where the model is uncertain:

$$r_{\text{robust}}(s, a) = r(s, a) - \lambda U_\eta(s, a).$$

This pushes the agent toward actions whose predicted outcomes are more reliable.

The same uncertainty estimate can encourage exploration or caution, depending on the objective.

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Exploration as Model Improvement

Exploration as Model Improvement

Exploration is key to learning a good policy.

In model-free RL, exploration often means trying actions that may lead to high reward.

In model-based RL, exploration can also mean trying actions that improve the model.

For example, an agent may choose an action because it helps reduce uncertainty about:

- the transition model,
- the reward model,
- or the consequences of future actions.

The value of an action includes both reward and information.

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Exploration as Model Improvement

A common exploration principle is optimism under uncertainty.

If the model is uncertain about (s, a), treat it as potentially valuable:

$$\tilde{R}(s, a) = \hat{R}(s, a) + \beta U(s, a),$$

where:

- $\hat{R}(s, a)$ is the estimated reward,
- $U(s, a)$ is model uncertainty,
- β controls exploration strength.

The agent is encouraged to visit uncertain state-action pairs.

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With optimism, the planner solves:

$$\pi = \arg\max_{\pi} \mathbb{E}_{\hat{M}} \left[\sum_{t=0}^{\infty} \gamma^t \left(\hat{r}(s_t, a_t) + \beta U(s_t, a_t) \right) \right].$$

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Planning with an Exploration Bonus

With optimism, the planner solves:

$$\pi = \arg\max_{\pi} \mathbb{E}_{\hat{M}} \left[\sum_{t=0}^{\infty} \gamma^t \left(\hat{r}(s_t, a_t) + \beta U(s_t, a_t) \right) \right].$$

The policy may choose an action because:

- it has high estimated reward, or
- it has high uncertainty and may improve the model.

The exploration bonus makes information-seeking part of the planning objective.

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Uncertainty-Aware Planning

Two Attitudes To Decision-making Under Uncertainty

Uncertainty can be treated in different ways.

Optimistic planning:

uncertain $\Rightarrow$ possibly useful.

Good for exploration.

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Two Attitudes To Decision-making Under Uncertainty

Uncertainty can be treated in different ways.

Optimistic planning:

uncertain $\Rightarrow$ possibly useful.

Good for exploration.

Conservative planning:

uncertain $\Rightarrow$ possibly dangerous.

Good for safety-critical domains.

Whether uncertainty is attractive or dangerous depends on the objective.

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Risk-Sensitive Model-Based Planning

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A risk-sensitive planner may penalize uncertain trajectories:

$$\pi = \arg\max_{\pi} \mathbb{E}_{\hat{M}} \left[\sum_t \gamma^t r_t \right] - \lambda \cdot \mathrm{Risk}(\pi, \hat{M}).$$

Risk-Sensitive Model-Based Planning

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A risk-sensitive planner may penalize uncertain trajectories:

$$\pi = \arg\max_{\pi} \mathbb{E}_{\hat{M}} \left[\sum_t \gamma^t r_t \right] - \lambda \cdot \mathrm{Risk}(\pi, \hat{M}).$$

Possible risk terms:

- probability of collision,
- variance of return,
- probability of entering unknown states,
- worst-case return over model ensemble members.

Suppose a robot compares two actions: take known route a_1, and enter unexplored route a_2.

Estimated rewards and uncertainty:

Action	$\hat{R}(s,a)$	$U(s,a)$
a_1 known route	6	1
a_2 unexplored corridor	7	5

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Example: Optimistic vs. Conservative

Suppose a robot compares two actions: take known route a_1, and enter unexplored route a_2.

Estimated rewards and uncertainty:

Action	$\hat{R}(s,a)$	$U(s,a)$
a_1 known route	6	1
a_2 unexplored corridor	7	5

$$\tilde{R}(s,a) = \hat{R}(s,a) + \beta U(s,a), \qquad \beta = 1.$$

$$\tilde{R}(s,a_1) = 6 + 1 \cdot 1 = 7, \qquad \tilde{R}(s,a_2) = 7 + 1 \cdot 5 = 12.$$

The optimistic planner chooses the unexplored corridor.

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Now suppose uncertainty is treated as risk.

The planner uses:

$$\tilde{R}(s,a) = \hat{R}(s,a) - \lambda U(s,a), \qquad \lambda = 1.$$

$$\tilde{R}(s,a_1) = 6 - 1 \cdot 1 = 5, \qquad \tilde{R}(s,a_2) = 7 - 1 \cdot 5 = 2.$$

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Now suppose uncertainty is treated as risk.

The planner uses:

$$\tilde{R}(s,a) = \hat{R}(s,a) - \lambda U(s,a), \qquad \lambda = 1.$$

$$\tilde{R}(s,a_1) = 6 - 1 \cdot 1 = 5, \qquad \tilde{R}(s,a_2) = 7 - 1 \cdot 5 = 2.$$

So:

$$\tilde{R}(s,a_1) > \tilde{R}(s,a_2).$$

The conservative planner chooses the known route, even though the unexplored path has a higher estimated reward.

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In this module, we moved from basic model-based RL to uncertainty-aware model-based decision making.

Main ideas:

- **Model learning:** learn a transition and reward model from experience.
- **Planning with learned models:** use the model to predict, simulate, and choose actions.
- **Model error:** learned models are imperfect, and errors can accumulate over rollouts.
- **Uncertainty estimation:** quantify where the model is reliable and where it is uncertain.
- **Exploration as model improvement:** choose actions that improve the model, not only actions with immediate reward.
- **Uncertainty-aware planning:** treat uncertainty as either an opportunity or a risk, depending on the objective.

What Have We Seen?

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Model-based RL is powerful because the agent can reason about possible futures, but it must also reason about the reliability of its own model.

Caution: Reward Shaping

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Reward shaping means modifying the reward signal in order to guide learning or planning.

Instead of optimizing the original task reward

$$R(s, a),$$

we optimize a shaped reward

$$\tilde{R}(s, a) = R(s, a) + F(s, a, s'),$$

where F is an additional shaping term.

Caution: Reward Shaping

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Reward shaping can be very useful:

- it can make sparse rewards easier to learn from,
- it can encourage useful intermediate behavior,
- it can guide exploration toward promising regions.

Reward shaping gives the agent extra guidance beyond the original task reward.

Caution: Reward Shaping

Reward shaping must be used carefully, because it can change the problem the agent is actually solving.

Possible challenges include:

- **Changing the optimal value:** the shaped reward may make a different behavior look optimal.
- **Scale mismatch:** the shaping term may dominate the real task reward.
- **Reward hacking:** the agent may exploit the shaping reward without solving the task.
- **Poor generalization:** behavior encouraged by the shaping term may not help in new situations.
- **Tuning difficulty:** the effect depends strongly on the weight of the shaping term.

A shaped reward changes the learning signal.

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